

Project Proposal

Group 30

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1 Overview

This project focuses on segmenting adult gliomas from MRI scans using a 2D U-Net convolutional neural network. The BraTS 2021 dataset provides four co-registered MRI sequences for each subject: T1, T1CE, T2, and T2-FLAIR, together with an expert-annotated 3D tumor segmentation mask. Our model will operate on 2D axial slices extracted from these volumes. For each slice, the corresponding images from all four MRI sequences are stacked as input channels, and the model predicts which image regions correspond to background or to one of the tumor sub-regions defined by BraTS.

Manual tumor segmentation requires trained raters to delineate tumor regions throughout a 3D MRI volume and is therefore time-consuming. An automated model can generate an initial segmentation that can be reviewed and corrected by an expert, reducing the amount of manual delineation required. Automated segmentation is also useful for quantitative research because a fixed model applies the same learned segmentation procedure to every scan, providing more repeatable measurements across large datasets than repeated manual segmentation. This can make it easier to compare tumor regions across subjects or across scans collected at different time points.

Brain tumor segmentation is challenging because gliomas vary substantially between subjects in size, shape, location, and appearance. Tumor tissue can also resemble surrounding healthy tissue, and tumor regions may occupy only a small fraction of an MRI slice, creating a strong imbalance between tumor and background. In addition, different tumor sub-regions are distinguished by patterns across T1, T1CE, T2, and T2-FLAIR rather than by a single MRI sequence alone. The model must therefore combine information from the four aligned inputs while identifying irregular tumor boundaries and separating tissue classes with similar appearances.

2 Task Definition

2.1 Machine Learning Task

The project is a supervised semantic segmentation task. Rather than assigning one class to an entire MRI slice, the model performs dense multiclass classification by assigning a class to each spatial location in the slice. The ground-truth labels are provided by the expert-annotated segmentation masks included with the BraTS 2021 dataset. Each location is assigned exactly one class, making this a single-label, four-class segmentation problem rather than a multi-label classification problem.

2.2 Model Input

Each model input consists of four corresponding axial MRI slices from the same subject and anatomical position. The slices are taken from the T1, T1CE, T2, and T2-FLAIR volumes, which have already been co-registered so that the same spatial location corresponds to the same anatomy in each sequence. The four 240×240 slices are stacked as separate input channels, producing a $240 \times 240 \times 4$ input to the 2D U-Net.

Using all four sequences allows the model to use differences in tissue appearance across the MRI sequences when distinguishing tumor sub-regions. For example, T1CE provides information about contrast-enhancing tissue, while T2 and T2-FLAIR provide different contrast for tissue with increased water content, including regions associated with edema.

2.3 Model Output and Segmentation Classes

For each input slice, the model produces a 240×240 segmentation map with one predicted class for each spatial location. The original BraTS 2021 segmentation masks use labels 0, 1, 2, and 4. Before training, label 4 will be remapped to 3 so that the model predicts four consecutive classes:

- Class 0: background / non-tumor tissue
- Class 1: necrotic and non-enhancing tumor core (NCR/NET)
- Class 2: peritumoral edema / invaded tissue (ED)
- Class 3: enhancing tumor (ET), corresponding to label 4 in the original BraTS masks

The predicted 2D segmentation therefore corresponds directly to the labels from the matching axial slice of the original 3D BraTS segmentation mask.

2.4 Technical Challenges

The segmentation task is difficult because the four classes are highly imbalanced: background tissue occupies most of an MRI slice, while individual tumor sub-regions may contain relatively few pixels. Gliomas also vary substantially between subjects in their size, shape, location, and appearance, so the model must generalize across tumors with very different morphologies. Boundaries between tumor sub-regions may be irregular or visually ambiguous, requiring the model to distinguish tissue classes using

information distributed across the four MRI sequences. Finally, because the proposed model operates on individual 2D slices, it does not directly use information from adjacent slices in the 3D MRI volume.

3 Data Source(s) and Plan for Data Collection

3.1 yashika TODO

open data points example in itk snap and confirm slice indices

3.2 Primary Data Source

The primary data source is the **RSNA-ASNR-MICCAI Brain Tumor Segmentation (BraTS) 2021 Challenge dataset** (Task 1: segmentation). It is the standard multi-institutional, expert-annotated multi-modal MRI benchmark for glioma segmentation. Each subject provides four co-registered MRI volumes (T1, T1CE, T2, T2-FLAIR) and an expert segmentation mask, which matches our design of stacking the four modalities as input channels to a 2D U-Net and predicting pixel-wise tumor sub-region labels.

3.3 Official Sources and Access

- **Primary challenge platform (Synapse / Sage Bionetworks):**

<https://www.synapse.org/#!Synapse:syn51514105> (training data under syn51514105 or successor links). The project uses the BraTS adult glioma MRI dataset, originally associated with the BraTS 2020/2021 challenges. The original BraTS 2021 challenge is now closed and its dataset is no longer directly hosted through the original challenge page. According to the BraTS organizers, the BraTS-GLI dataset made available through the BraTS 2023 challenge contains the same BraTS 2020/2021 data, with the enhancing tumor (ET) label changed from 4 to 3.

- **The Cancer Imaging Archive (TCIA) mirror / original contributing collections:** DOI <https://doi.org/10.7937/jc8x-9874> (RSNA-ASNR-MICCAI-BraTS-2021). This hosts the processed NIfTI volumes (and some DICOM) for approximately 1480 subjects of the challenge cohort under a CC-BY 4.0 license. Underlying raw DICOM from contributing collections (TCGA-GBM, TCGA-LGG, IvyGAP, CPTAC-GBM, UPENN-GBM, ACRIN-FMISO-Brain, UCSF-PDGM, etc.) are

available via TCIA’s NBIA Data Retriever under the respective collection licenses (mostly open or controlled-access with DUA).

- **Kaggle** (convenient for download, but secondary): example <https://www.kaggle.com/datasets/dschettler8845/brats-2021-task1> Related uploads such as [awsaf49/brats2020-training-data](https://www.kaggle.com/datasets/awsaf49/brats2020-training-data) for earlier years are available.

3.4 Terms of Service, Licensing and Accessing the Data

- **No web scraping is performed.** The data is a formally published research dataset obtained through official portals (Synapse client, TCIA download tools) or the Kaggle mirror. There is no live API polling, so rate limiting is not a concern. Large downloads use the authenticated bulk-download tools provided by these platforms.
- **License:** released for research use with attribution (CC-BY 4.0 on TCIA). We will use the data for non-commercial academic purposes only and will not attempt to re-identify subjects. All data was de-identified by the organizers before release.
- **Required acknowledgement:** we will include the acknowledgement statement requested by the organizers, referencing the RSNA-ASNR-MICCAI BraTS Challenge and Synapse ID syn25829067 and cite the papers listed in the Citations subsection.

3.5 How the Data Were Collected, Annotated and Pre-processed

Acquisition: Retrospective multi-institutional routine clinical pre-operative multiparametric MRI (mpMRI) of adult gliomas (WHO grades II–IV, predominantly glioblastoma / high-grade and lower-grade gliomas). Scanners and protocols vary across approximately 15 institutions (United States, Europe, India, etc.), producing realistic heterogeneity in field strength (1.5T / 3T), sequence parameters and image quality.

Modalities: native T1, post-contrast T1-weighted (T1Gd / T1CE), T2-weighted and T2-FLAIR. Each subject contributes four co-registered 3-D volumes.

Annotation protocol: Labels were produced by manual delineation by 1 - 4 trained raters following a standardized BraTS protocol, then reviewed and approved by experienced neuroradiologists. **We do not label any data ourselves**, so no per-instance annotation time applies. (Manual voxel-level delineation of

a single 3D glioma volume is a labor-intensive expert task, which is precisely why using a pre-annotated benchmark is appropriate.) The mask labels are:

Label	Region	Abbrev.
0	Background / non-tumor	—
1	Necrotic and non-enhancing tumor core	NCR/NET
2	Peritumoral edema / invaded tissue	ED
4	GD-enhancing tumor	ET

Standard evaluation regions are Whole Tumor = $\{1, 2, 4\}$, Tumor Core = $\{1, 4\}$, and Enhancing Tumor = $\{4\}$. Label 3 does not exist in BraTS masks. Before computing the loss we will remap $4 \rightarrow 3$, giving a 4-class problem with labels $\{0, 1, 2, 3\}$.

Preprocessing applied by the organizers: DICOM to NIfTI conversion (with PHI removed), rigid co-registration of the four modalities, registration to the SRI24 anatomical template, resampling to 1 mm^3 isotropic resolution, and skull-stripping. Volumes are $240 \times 240 \times 155$ voxels stored as `.nii.gz`.

Preprocessing performed by us: Intensities have no universal scale in MRI, so we will apply per-volume, per-modality z-score normalization computed over non-zero (brain) voxels only, optionally with percentile clipping. We will then stack the four modalities as channels, extract axial 2D slices, remap the labels as above, and apply training-time augmentation (flips, small rotations, elastic deformation, intensity shifts) applied identically across channels and mask.

3.6 Available Metadata

Each case is identified by a subject ID (BraTS2021_XXXXX). The release also provides, for subsets of cases, a mapping to original TCIA/TCGA identifiers and MGMT promoter methylation status (used in the radiogenomic Task 2). Metadata is not used as a model input in this project the model input is the four imaging modalities only.

3.7 Dataset Size and Use of the Data

- **3D volumes:** 1251 publicly labeled training subjects. Each contributes four modality volumes and one mask. A further 219 validation cases are released with hidden labels and cannot be used for local evaluation, so all train/validation/test splits are drawn from the 1251 labeled subjects.

- **2D slices:** each subject has 155 axial slices, giving roughly 194,000 slices in total. After discarding empty (all-zero) slices and subsampling slices without tumor, we expect on the order of tens of thousands of multi-channel training slices. The exact count will be computed after download. This far exceeds the 1000-data-point minimum.
- **Entire dataset vs subset:** we plan to use the *entire* labeled cohort. Tumor pixels are a small minority of each slice, so every tumor-containing slice is valuable and inter-subject variability in tumor appearance and scanner characteristics supports generalization. If memory or compute limits arise during prototyping, a random subject-level subset (30–50%) may be used temporarily, with final models trained on the full set.
- **Splitting:** the train/validation/test split (approx. 70/15/15) is done *by subject*, never by slice, to prevent leakage between adjacent slices of the same patient.

Why the entire labeled training cohort rather than a small subset. The project goal is accurate segmentation under realistic appearance variation. Using the full 1251-subject set (and the derived 2-D slices) maximizes statistical power, captures the documented inter-subject heterogeneity of tumor morphology and intensity and permits rigorous evaluation of the 2-D U-Net against published BraTS benchmarks. Computational resources permitting, the complete set will be used. If memory constraints arise during early prototyping, a random but stratified 30 - 50 % subject subset can be employed temporarily, with final models re-trained on the full set.

3.8 Example Data Points

Each “data point” for the 2-D U-Net is a 4-channel 240×240 (or cropped/resized) axial slice together with its corresponding integer label map.

1. **Subject BraTS2021_00000, axial slice $\approx z = 78$** Input: stacked [T1, T1CE, T2, FLAIR] intensities (skull-stripped, co-registered). Label map: predominantly background (0) with a central necrotic core (1), surrounding enhancing rim (4) and peripheral edema (2). Typical whole-tumor Dice target region occupies $\sim 5 - 15\%$ of the slice.
2. **Subject BraTS2021_00123, axial slice $\approx z = 95$** Input: same four modalities. Label map: larger infiltrative edema (2) with smaller non-enhancing core (1) and sparse enhancing foci (4). Demon-

strates the visual similarity of tumor tissue to surrounding healthy white matter that makes the task challenging.

3. **Subject BraTS2021_00456, axial slice $\approx z = 62$** Input: four modalities. Label map: almost pure enhancing tumor (4) with minimal necrosis, illustrating inter-subject morphological variability. Background class still dominates the pixel count, motivating Dice / focal / class-weighted losses.

3.9 Citations

The following works are cited for use of this dataset:

1. U. Baid et al., “The RSNA-ASNR-MICCAI BraTS 2021 Benchmark on Brain Tumor Segmentation and Radiogenomic Classification,” arXiv:2107.02314, 2021.
2. B. H. Menze et al., “The Multimodal Brain Tumor Image Segmentation Benchmark (BRATS),” *IEEE Trans. Med. Imaging*, 34(10):1993–2024, 2015.
3. S. Bakas et al., “Advancing The Cancer Genome Atlas glioma MRI collections with expert segmentation labels and radiomic features,” *Scientific Data*, 4:170117, 2017.
4. Kaggle mirror: <https://www.kaggle.com/datasets/dschettler8845/brats-2021-task1>.

4 Solution

U-Net is a convolutional neural network (CNN) architecture designed for image segmentation tasks. This architecture consists of an encoder and decoder. The encoder repeatedly applies convolutions and downsamples the image. It learns increasingly abstract features. The decoder upsamples these features back up to the original resolution of the input. Its job is to figure out the exact location of features. In the context of our project, encoded early layers will detect edges and features, while deeper layers can learn patterns associated with tumors. [1]

Each MRI scan consists of four images: T1, contrast-enhanced T1 (T1CE), T2, and T2-FLAIR. These images will be stacked as separate input channels, allowing the model to use information from each image concurrently in its prediction.

The target labels will consist of the already labeled segmentation masks provided with the dataset. Therefore, as the labels are pixel level, the model will also perform pixel-wise classification, assigning

each pixel in the U-Net slice to either the background or one of the annotated tumor regions. As such, the model will be trained as a 2D multiclass U-Net on axial slices.

Each MRI modality will be normalized separately since MRI intensities do not have a universal scale. 2D axial slices will be extracted, and T1/T1CE/T2/FLAIR will be stacked as four channels. Augmentation will then be applied during training.

The split will be done by patient/subject, not by individual slice. Otherwise adjacent slices from the same patient could appear in both training and testing which will artificially inflate results.

Dice score will be used as the main segmentation metric. It will be evaluated for individual tumor regions. We will compare cross-entropy loss versus Dice-based loss or a combined Dice + cross-entropy loss.

Relevant libraries are PyTorch, NumPy, NiBabel (for reading nii.gz data files).

References

- [1] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. *CoRR*, abs/1505.04597, 2015.